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A LAB REPORT ON THE RC TIME CONSTANT

1 Theory

An electrical circuit consisting of a resistor and a capacitor—commonly referred to as an RC circuit—exhibits a fundamentally different behavior from a purely resistive circuit. At the heart of this difference is the capacitor’s ability to store charge on its plates. For a capacitor subjected to an electric potential difference V , the relationship between the stored charge Q and the voltage is given by:

$$Q = CV$$

where C is the capacitance, a property of the capacitor that quantifies how much charge it can store per unit voltage. What makes RC circuits particularly interesting is not just the fact that charge accumulates, but *how* it accumulates over time.

When a switch is closed and the capacitor begins to charge, the rate at which charge builds up on the plates is not constant—it slows down as the voltage across the capacitor approaches that of the power supply. This behavior is characteristic of an exponential function, a special class of function whose rate of change is proportional to its current value. In other words, the more charge the capacitor has already accumulated, the slower it continues to charge. This same logic applies in reverse during discharging: the more charge remains, the faster it leaves, and the process slows as the plates approach neutrality.

To see where this exponential form comes from, we can apply Kirchhoff’s voltage law to the RC circuit during charging. The voltage supplied by the source $\mathcal{E}$ must equal the sum of the voltage drops across the resistor and the capacitor:

$$\mathcal{E} = V_R + V_C = IR + \frac{Q}{C}$$

Since the current is the rate of change of charge, $I = dQ/dt$, we can substitute directly into the equation above to obtain the following first-order differential equation:

$$\mathcal{E} = R \frac{dQ}{dt} + \frac{Q}{C}$$

Rearranging to isolate the derivative:

$$\frac{dQ}{dt} = \frac{\mathcal{E}}{R} - \frac{Q}{RC}$$

This equation tells us that the rate at which charge accumulates on the capacitor decreases as Q grows—precisely the behavior we described as exponential. Solving this differential equation with the initial condition $Q(0) = 0$, and converting back to voltage using $V_C = Q/C$, yields the solution for the voltage across the capacitor during charging:

$$V_C(t) = \mathcal{E} (1 - e^{-t/\tau})$$

and during discharging from an initial voltage V_0 :

$$V_C(t) = V_0 e^{-t/\tau}$$

In both expressions, τ is the **RC time constant**, defined as:

$$\tau = RC$$

The time constant τ carries units of seconds and has a concrete physical meaning: it is the time it takes for the capacitor to charge to approximately 63.2% of the supply voltage, or to discharge to approximately 36.8% of its initial voltage—values that follow directly from evaluating e^{-1} .

The goal of this experiment is to verify this theoretical prediction experimentally using an RC circuit built on a breadboard with a known resistance $R = 3,300\ \Omega$ and capacitance $C = 470\ \mu\text{F}$, giving a theoretical time constant of:

$$\tau_{\text{theory}} = RC = (3,300)(470 \times 10^{-6}) \approx 1.551\ \text{s}$$

Rather than measuring charge or current directly, we record the voltage across the capacitor over time using the PASCO 850 interface and voltage sensor, as voltage is the most directly accessible quantity given our available instrumentation.